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Use of Stochastic Dynamic Programming for Reservoir Management

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Hydropower production plays an important role in the operation of electric energy production-distribution systems. Since most of the economically feasible hydroelectric sites have already been developed, it is necessary to examine and develop practical, real-time operational models which can be used to increase the output from existing hydropower plants. Three main issues are addressed by this research: the potential of increasing the output from existing hydropower plants, the alleviation of dimensionality problems for multistate dynamic programming, and the use of probabilistic forecast in the decision-making process. An optimization model is developed which can be used as an analytical tool in the decision-making process for reservoir operation. The model takes into consideration the uncertainty of forecast at the time a policy must be established. The uncertainty is expressed in terms of the second moments of the forecast probability distributions. There is no limitation on the type of distribution, and it is assumed that forecast is made by a conceptual type of watershed model. The proposed methodology is applicable to constrained stochastic systems with quadratic objective functions and linear dynamics. It uses the decomposition principle of dynamic programming without discretizing the state or control variable and therefore the method can be used for large-scale systems. It is an iterative procedure which requires an initially feasible solution and solves a series of quadratic programming problems at each iteration. The applicability of the research is demonstrated through case studies.

INTRODUCTION

A considerable amount of research has been done in the area of reservoir operation. A recent paper by *Yeh* [1985] reviewed the state-of-the-art of management models developed for reservoir operations. Much of the research reported in the area of optimization for multireservoir system operation has been based on the fundamental assumption that future events are known with certainty. Because of the problem of dimensionality, the research in the area of stochastic optimization is usually limited by the number of reservoirs.

The main problem with the optimization of large-scale systems is the "curse of dimensionality." The problem is aggravated in the case of stochastic optimization. In the area of reservoir management, dynamic programming (DP), as originally developed by *Bellman* [1957], requires discretizing the state and control variables. In order to alleviate the dimensionality problem associated with the discrete DP, one has to develop a methodology by which storage requirements on a modern computer would not expand exponentially as the size of the problem increases. Under certain conditions, one can use the concept of DP without discretizing the state or control variable, i.e., differential DP [*Jacobsen and Mayne*, 1970]. *Murray and Yakowitz* [1979] used this concept in a deterministic framework. The proposed research extends the use of differential DP to a reservoir management problem where inflows into the reservoirs are considered to be stochastic.

The production of the maximum possible energy from a system of hydro facilities is limited by the ability of making an accurate forecast of future streamflows. Existing models dealing with streamflow forecast are not perfect, and their output is only a statistical approximation of the actual event. We assume that all modeling uncertainties can be successfully described by the first and second moments of the underlying probability distribution.

Mathematical models based on a monthly time period cannot capture the system dynamic response that occurs in time intervals shorter than a month. For a hydro-reservoir system, the water balance equation of its monthly model should incorporate the fact that spill may occur during the month, although average storage is below capacity. Naturally, the higher the storage, the higher the spill. For a deterministic model, spill could have been considered as an increasing function of storage. For a stochastic model a more realistic consideration would be to express spill as a dependent random variable.

For the purpose of this research we define spill as the amount of water that enters the reservoir, but exits without passing through the penstocks. Therefore under this general definition, spill may include spillway releases, releases through the bypasses, evaporation, groundwater seepage, etc.

Since we consider that the system is a stochastic one, we assume that constraints require the expected values to be within specified limits. Another approach could have been to use chance constraint formulation. Essentially the two approaches ("expected value" or "chance constraint") are similar. They both require the conversion of the stochastic constraints to their deterministic counterparts.

The direct, or immediate, benefit from the application of this research would be the increase of hydropower production by simply operating the existing facilities more efficiently [*U.S. Department of Energy, Energy Information Administration*, 1983*a*, *b*, 1985; *Stanford Research Institute*, 1974; *Davis and Buckley*, 1984]. The indirect benefit is the development of a stochastic optimization methodology which can be applied to reservoir management and other areas of water resources and system analysis. The necessity for such development has explicitly been stated by *Grygier and Stedinger* [1985].

PROBLEM FORMULATION

Framework Overview

The primary objective of the problem addressed herein is to maximize the total energy produced by an existing system of hydroelectric facilities but subject to typical system con-

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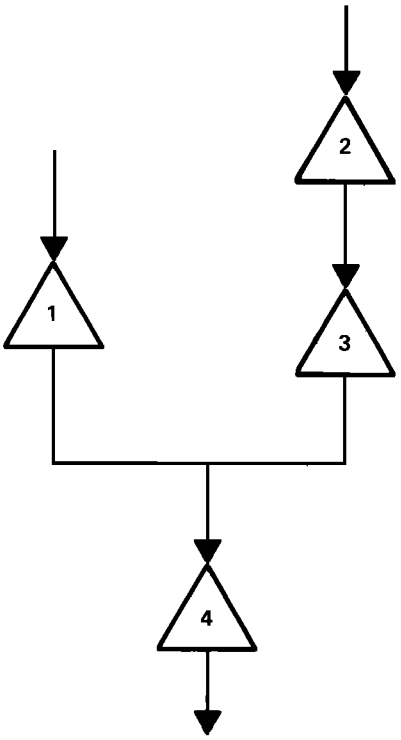


Fig. 1. Example of reservoir system representation.

straints. Our approach is based upon the following observation: hydroelectric production during a certain period of time is directly proportional to the product of the total productive release times the average operating head. Assume that a given reservoir has never spilled during a specific operation. Then, a marginal increase in operating head would result in an increased hydropower production. Any solution in which spill is an impossible event is suboptimal to the solution in which spill is marginally feasible. Therefore the optimum operation should balance the losses due to spill during the optimization horizon versus the gain of maintaining the reservoir pool at the highest possible level.

There are only natural inflows into the system which are stored in the reservoirs or routed through a chain of hydro facilities. The flow of water from one facility to another is one directional. Ultimately, the water flows out of the system. Future inflows are forecasted for discrete time intervals of a certain length; the time period used herein is 1 month. For the basic formulation, forecasted inflow is uncorrelated from period to period, but it may be correlated from site to site. Therefore we have assumed cross-correlations among sites. However, we also discuss how we can extend the formulation to handle the case of serially correlated input. For cross-correlated input, the uncertainty of forecast can be expressed in terms of a covariance matrix for each time period of forecast.

The proposed research is applicable to a system of reservoirs of any configuration. The reservoirs can be in series or in parallel. One reservoir can provide water to more than one downstream reservoir and may receive water from more than one upstream reservoir. Figure 1 shows a typical reservoir system for which our model is applicable. The four reservoir system has been used by several researchers in the past as a standard case study [*Murry and Yakowitz*, 1979; *Ozden*, 1984; *Pereira and Pinto*, 1985].

In a compatible approach to *Becker and Yeh* [1974] we assume that certain contract levels have already been established. Our simplest formulation assumes that the hydro facilities have been contracted to the maximum deliverable on-peak energy and the model maximizes the secondary (off-peak) energy production. This is equivalent to the case where any additional energy produced beyond the contracted level is sold at a constant unit price. If the contract levels are below the maximum deliverable on-peak energy (as in the case of Figure 2 for periods 3–6), then given the relative worth of on-peak/off-peak energy the model will maximize the total benefit (value of energy) from hydropower operation.

Objective Function

On the basis of the previously established framework, the objective function of the proposed optimization model can be stated as

$$\max_{CV\ RV} E\left\{\sum_{t=1}^{NP} HA^T(t)\cdot R(t)+[S(NP+1)-TARG(NP+1)]^T \cdot PEN\cdot[S(NP+1)-TARG(NP+1)]\right\} \quad (1)$$

where

CV set of control variables;
RV set of random variables;
NP number of periods within the optimization horizon;
$E\{\ \}$ mathematical expectation;
T transpose of a vector;
R(t) power release during period (month) t, vector of size NS;
NS number of reservoirs in the system;
HA(t) average head elevation during period t, vector of size NS;
S(NP + 1) storage at the beginning of period NP + 1, vector of size NS;
TARG(NP + 1) target value for the ending storage at time NP, vector of size NS;
PEN (NS × NS) matrix expressing the penalty cost of deviating from the target values.

The last term in the above equation represens the penalty of deviating from the ending target value TARG(NP + 1). Simplifying this term with respect to S(NP + 1), and assuming,

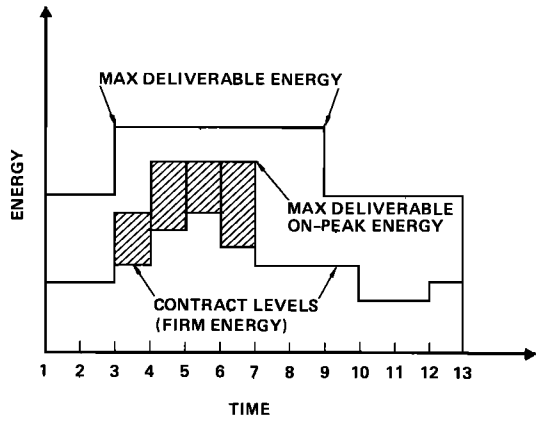


Fig. 2. Alternate on-peak energy generation.

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without loss of generality, that PEN is a symmetric matrix, we obtain

$$\max_{CV\ RV} E\left\{\sum_{t=1}^{NP} HA^T(t)\cdot R(t)+S^T(NP+1)\cdot PEN\cdot S(NP+1) - 2\cdot TARG^T(NP+1)\cdot PEN\cdot S(NP+1) + TARG^T(NP+1)\cdot PEN\cdot TARG(NP+1)\right\} \quad (2)$$

Constraints

There are several constraints necessary to formulate the system according to the established framework. The water mass balance is guaranteed by the system dynamic equations

$$S(t+1) = S(t) + IM(t) - SYS\cdot R(t) - SP(t) \quad t = 1, \cdots, NP \quad (3)$$

where

SYS (NS × NS) matrix describing the system configuration;
IM(t) reservoir inflow during period t, vector of size NS;
SP(t) water losses during period t, vector of size NS.

In the above equation, SP(t) represents the total amount of water that has failed to pass through penstocks. Its actual value is a random variable at the time an operational policy has to be established. IM(t) is an independent, with respect to time, multivariate random variable with mean QM(t) and covariance matrix QV(t). S(t) is the state variable at time period t, and it is also a dependent random variable.

We consider that the average head for the period t, HA(t), is sufficiently expressed as

$$HA(t+1) = \{H(t) + H(t+1)\}/2 \quad t = 1, \cdots, NP \quad (4)$$

where H(t) is the head at the beginning of period t.

Elevation and storage are related through a characteristic storage-elevation curve. In practice this curve is usually given as a continuous piece-wise linear function; therefore we can write

$$H(t+1) = A(t)\cdot S(t) + b(t) \quad t = 1, \cdots, NP \quad (5)$$

where A(t) is a diagonal matrix containing the coefficients of the linear relationship and b(t) is a vector of the constant terms.

Spill SP(t) is a random variable at the time a decision must be made. Based on the data we used for the case study, we have observed that the higher the storage, the higher the amount of spill. Moreover, spill is greater as inflow variance increases. Here we assume that spill can be expressed as

$$SP(t) = \alpha(t)\cdot S(t) + \beta(t)\cdot S(t+1) + \gamma(t)\cdot IM(t) \quad t = 1, \cdots, NP \quad (6a)$$

where $\alpha(t)$, $\beta(t)$, and $\gamma(t)$ are random coefficient matrices expressing the statistical characteristics of the system. The probabilistic characteristics of these matrices can be obtained from the historical data. If no historical data exists, these coefficients have to be initially estimated and recalibrated when operational data becomes available. Substituting the expression for S(t + 1) from (3) into (6*a*), the spill can be expressed as

$$SP(t) = [I + \beta(t)]^{-1}\cdot[\alpha(t) + \beta(t)]\cdot S(t) + [I + \beta(t)]^{-1} \cdot [\beta(t) + \gamma(t)]\cdot IM(t) - [I + \beta(t)]^{-1}\cdot\beta(t)\cdot SYS\cdot R(t) \quad (6b)$$

where I is the identity matrix. Equation (6*b*) shows that the decision variable, the amount of water released R(t), has a direct impact on the amount of water spilled.

Equations (3)–(6) are stochastic constraints. We call a constraint stochastic if it contains at least one random variable. The actual value of the right- or left-hand side of a stochastic constraint is not known at the time of the decision. Moreover, operational requirements provide boundary constraints, which, as stated previously, are imposed upon the mean values of the random variables. These constraints can be written as

Storage

$$SLL(t) \le E\{S(t)\} \le SUL(t) \quad t = 1, \cdots, NP \quad (7)$$

Release

$$RLL(t) \le R(t) \le RUL(t) \quad t = 1, \cdots, NP \quad (8)$$

Production

$$CONTR(t) \le E\{HA^T(t)\cdot R(t)\} \le CAPM(t) \quad t = 1, \cdots, NP \quad (9)$$

where SLL(t) and RLL(t) and SUL(t) and RUL(t) are the lower and upper bounds for storage and releases, respectively, while CONTR(t) and CAPM(t) are the contracted minimum and maximum hydropower production for the system, respectively. Equation (8) is a deterministic constraint. It contains the control vector R(t) and given constant parameters. Equations (7) and (9) constrain the means of dependent random variables. By applying the expectation operator $E\{\ \}$, these constraints can be converted to deterministic constraints on the control variables.

In the above optimization model the decision variables are the releases during each period, R(t), t = 1, ···, NP. For a DP formulation, state variables are the reservoir storages at the beginning of each period, S(t), t = 1, ···, NP + 1. These storage values represent the target values for the model next in the echelon (a weekly or daily model). The actual storage at the beginning of each period, which will be observed in the future is a random variable.

Constraints on storage and production could also involve probabilistic requirements, i.e., probability of production during a period to exceed a minimum acceptable amount. Then the so called chance constraints must be again converted to a set of "deterministic equivalents" using the assumed distribution of the random variables.

On-Peak Energy Consideration

The need to differentiate between on- and off-peak production arises from the fact that hydropower energy is worth more if it is produced during peak hours (prime time) rather than off-peak hours. Figure 2 illustrates the concept of contract levels, maximum deliverable on-peak energy, and maximum energy produced. For a hydro system, a pattern of on-peak hours for each month is usually established. The number of on-peak hours is determined by the demand for electricity and is independent of the hydrology. The maximum deliverable on-peak energy that can be produced by the system

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during a given period is equal to the product of the total number of on-peak hours for that period times the available system power capacity. Similarly, there are different contract levels for each period.

In the given formulation (equations (2)–(9)) it was assumed that contract levels were set at the maximum deliverable on-peak energy levels for every energy period. Implicitly, we have presented a formulation that maximizes off-peak production given that on-peak (contracted) demand is met. If this is not true, then for some periods the established contract levels are below the maximum deliverable on-peak energy. Therefore it will be beneficial to produce more (on-peak) energy during these months before attempting to maximize secondary (off-peak) energy production.

To quantify the above concept and successfully model the system, one needs to know the established contract levels for each period, the maximum deliverable on-peak energy, and the relative worth of on- and off-peak hydropower production. In the following sections we will demonstrate how to generalize the above formulation and take into consideration possible deviations of contract levels and maximum deliverable on-peak energy.

A rate structure with on-off peak pricing is assumed, as was done by *Grygier and Stedinger* [1985]. Let p(t) be the ratio of the worths of on-peak hydroelectric energy to the off-peak equivalent for period t. Also, let PEAK(t) be the maximum deliverable on-peak energy for period t. The objective (2) can then be written as

$$\max_{CV} E_{RV}\left\{\sum_{t=p}^{NP} HA^T(t)\cdot R_0(t) + p(t)\cdot HA^T(t)\cdot R_p(t) + S^T(NP+1)\cdot PEN\cdot S(NP+1) - 2\cdot TARG^T(NP+1)\cdot S(NP+1) + TARG^T(NP+1)\cdot PEN\cdot TARG(NP+1)\right\} \quad (10)$$

where $R_p(t)$ are the releases above the contract level CONTR(t) and are below the releases corresponding to the maximum deliverable on-peak energy PEAK(t). $R_0(t)$ are the off-peak hour releases.

The constraint set should then also include the following:

$$R(t) = R_0(t) + R_p(t) \qquad t = 1, \cdots, NP \quad (11)$$

$$E\{HA(t)^T\cdot R_p(t)\} \le PEAK(t) \qquad t = 1, \cdots, NP \quad (12)$$

PROPOSED PROCEDURE

The problem of maximizing the hydropower production as formulated in the previous section is a stochastic nonlinear optimization problem, since inflows during each time period are considered to be random variables and are described by the first and second moment of a multivariate probability distribution. Consequently, the expectation function is used to produce a deterministic equivalent. Usually, the derivation of a determinisitc equivalent produces a nonlinear expression. Moreover, even if the variance could have been neglected, the problem would have been nonlinear. The objective function involves the product of two variables, HA(t) and R(t), and a square term, the penalty function; therefore it is nonlinear. Also, the constraint set is nonlinear. The storage-elevation curve can be considered as piece-wise linear. The production bounding constraint (9) is also nonlinear.

There is no straightforward mathematical technique that can be applied directly to solve the above optimization problem. We propose a solution procedure analogous to the constrained differential DP [*Murray and Yakowitz*, 1979]. An extensive review of several DP models has been done by *Yakowitz* [1982]. Related research work includes *Goulter and Tai* [1985], *Mariño and Loaiciga* [1985], *U.S. Army Corps of Engineers* [1981], *Becker et al.* [1976], *Wasimi and Kitanidis* [1983], and *Stedinger et al.* [1984].

Gjelsvik [1982] presented a stochastic seasonal planning model for multireservoir, hydroelectric power systems by using differential DP. The objective function of his model minimizes the expected value of the variable operating costs (fuel costs of alternative energy resources) for a power company which owns the reservoir system and other power plant facilities. The model, although seasonal, neglects water head variations. It is also restricted to the case in which all power stations are feeding into a concentrated network represented by a single busbar.

Ikura and Gross [1984] examined the problem of large-scale hydro system scheduling with, what they called "forced" spill conditions. They formulated the problem as an optimization with a nonlinear objective function subject to linear constraints plus a set of nonlinear constraints resulting from the forced spill conditions. In the solution, the nonlinear constraints were approximated by piece-wise linear functions by appending the nonlinear terms to the objective function as a penalty term via the exact penalty method. The nonlinearity emanates from the assumption that there is no spill within a given period if the ending storage is below the maximum allowed level. The authors differentiated between forced and "controllable" spill. In contrast, we have assumed that spill is always uncontrollable; one does not make any release without passing it through the turbines unless he is forced to do so. However, as we indicated above, an operating policy that has spill as an unforeseen event cannot be optimal. Moreover, because in the modeling process we discretize a continuous, dynamic event, in order to preserve the statistical system characteristics, we assumed that spill is a random variable.

Stochastic Dynamic Programming

The proposed algorithm uses a DP approach to optimize the problem of hydropower production. The algorithm follows a concept similar to differential DP in the state and control variables are not discretized in the solution process. Differential DP can be applied successfully to optimization problems with linear dynamics and quadratic performance criterion functions [*Dreyfus and Law*, 1977]. In the most general case, this approach requires no bounds on the control or state variables and, in the case of minimization problems, a convex objective function. Differential DP is a successive approximation technique, based on DP rather than the calculus of variation, for determining optimal control of nonlinear systems [*Jacobson and Mayne*, 1970]. *Murray and Yakowitz* [1979] were the first to demonstrate the use of constrained differential DP in water resources problems.

Unconstrained differential DP can be applied to a system that is subject to linear or nonlinear dynamics. It requires the selection of an arbitrary initial policy and the "quadratization" of the objective function. If the objective function is of an order higher than that of quadratic Taylor series approximation around the initial estimate can be used to quadratize the function. If the objective is linear, by adding a small qua-

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dratic term, the quadratization is again achieved within tolerable limits. Moreover, if the dynamics are not linear, they must be approximated by linear expressions. *Murray and Yakowitz* [1979] noticed that the imposition of constraints upon the control or state variables does not cause insuperable problems inasmuch as an initial estimate is required anyway. They proposed quadratic programming to solve the constrained problem in a successive approximation procedure.

The relationship of our research with the work of *Murray and Yakowitz* [1979] is that we propose a similar solution procedure while assuming that the forecast is not perfectly known. We examine the application of constrained and stochastic DP in the area of reservoir management in order to increase hydropower production.

The main advantage of the proposed approach is its applicability to a system of several reservoirs while forecast is treated as uncertain system input. It avoids the dimensionality problems often associated with the use of conventional DP.

Theoretical Basis

By substituting the expression for spill, (6*a*), into the dynamic equation (3), we obtain

$$S(t+1) = S(t) + IM(t) - SYS \cdot R(t) - \alpha(t) \cdot S(t) - \beta(t) \cdot S(t+1) - \gamma(t) \cdot IM(t)$$

or

$$[I + \beta(t)] \cdot S(t+1) = [I - \alpha(t)] \cdot S(t) + [I - \gamma(t)] \cdot IM(t) - SYS \cdot R(t)$$

where I is the identity matrix. Premultiplying the last equation by $[I + \beta(t)]^{-1}$ we have

$$S(t+1) = [I + \beta(t)]^{-1} \cdot [I - \alpha(t)] \cdot S(t) + [I + \beta(t)]^{-1} \cdot [I - \gamma(t)] \cdot IM(t) - [I + \beta(t)]^{-1} \cdot SYS \cdot R(t) \qquad (13)$$

$$t = 1, \cdots, NP$$

Substituting (5) into (4), we have

$$HA(t) = \{[A(t) \cdot S(t) + \beta(t) + A(t+1) \cdot S(t+1) + \beta(t+1)\}/2 \qquad (14)$$

Finally, substituting the expression for $S(t+1)$ from (13) into (14), we obtain

$$HA(t) = \tfrac{1}{2} \cdot \{[A(t) + A(t+1) \cdot [I + \beta(t)]^{-1} \cdot [I - \alpha(t)]] \cdot S(t) - A(t+1) \cdot [I + \beta(t)]^{-1} \cdot SYS \cdot R(t) + A(t+1) \cdot [I + \beta(t)]^{-1} \cdot [I - \gamma(t)] \cdot IM(t) + b(t) + b(t+1)\} \qquad (15)$$

Using the above developed expression for the average head (15) and for the objective function (2), we derive

$$\max_{CV} \underset{RV}{E} \left\{ \frac{1}{2} \cdot \sum_{t=1}^{NP} S^T(t) \cdot [A(t) + [I - \alpha(t)] \cdot [I + \beta(t)]^{-1} A(t+1)] \cdot R(t) + R^T \cdot SYS^T \cdot [I + \beta(t)]^{-1} \cdot A(t+1) \cdot R(t) + [IM^T(t) \cdot [I - \gamma(t)] \cdot [I + \beta(t)]^{1} \cdot A(t+1) + b^T(t) + b^T(t+1)] \cdot R(t) + S^T(NP+1) \cdot PEN \cdot S(NP+1) - 2 \cdot TARG^T(NP+1) \cdot PEN \cdot S(NP+1) + z \right\} \qquad (16)$$

where z is a constant which is equal to $TARG^T(NP+1) \cdot PEN \cdot TARG(NP+1)$. Each term of the summation in the above form of the objective function (16) contains terms up to the quadratic order of the state variable $S(t)$, $t = 1, \cdots, NP+1$ and control variable $R(t)$, $t = 1, \cdots, NP$. State and control variables are related through the dynamic equation (13). The optimization of the objective function (16) is subject to the system of linear dynamics (13) plus the boundary constraints (equations (7) through (9)).

In the absence of the boundary constraints the optimization problem can be solved by DP as a stochastic problem with linear dynamics and quadratic criteria if, and only if, the recursive equation is a concave function of the state variables [*Dreyfus and Law*, 1977]. The concavity, although it cannot be guaranteed, seems to be valid for the given formulation of hydropower systems. A formal general proof of the concavity of the recursive equation cannot be given because the latter depends upon the specific system characteristics. Here it has been assumed that (6*a*) sufficiently describes the system characteristics. The actual shape of the objective function (16) and, consequently, the concavity of the recursive equation depends upon the specific expression for the spill. Under the postulation that (6*a*) is general enough, the concavity depends on the specific values of the coefficient matrices $\alpha(t)$, $\beta(t)$, and $\gamma(t)$.

On the basis of our experience with the model's performance, an intuitive explanation of the desirable and achievable concavity is that if the storage level is lower than optimal, then it pays to increase the storage, since the increase of the objective function due to higher head is greater than the negative contribution of the increased amount of spill. Increasing the storage level beyond optimal makes the energy losses due to spill greater than the additional amount of energy that can be produced due to higher operating head.

The proposed DP approach takes advantage of the fact that the objective function is quadratic and is subject to linear dynamics. In such a case, at each stage of the DP, the recursive equation is a quadratic function of the state variable. If there are no constraints other than the linear dynamics, the recursive equation can be computed analytically by solving an unconstrained quadratic optimization problem and the solution can proceed backwards. In the case of a stochastic formulation, if the goal is to optimize the objective function, i.e., its expectation, stochasticity does not cause an insuperable situation. The uncertainty of future events can be handled by applying the expectation operator on the random terms of the quadratic objective. If the term is quadratic, its expectation is equal to the square of its mean value plus its variance. If the term is linear, its expectation is equal to its mean.

With or without considering uncertainty, in the presence of the bounding constraints, one could proceed by examining whether each constraint is active or not. In such a case, the computational and storage requirements would grow exponentially with the dimension of the control and state space and, consequently, one of the main advantages of a DP approach that avoids discretizing the state and control variables would be lost. In the presence of bounding constraints, an initial solution can be improved by a constrained quadratic optimization at each stage in lieu of the unconstrained, quadratic optimization.

With no bounding constraints, during the backward solution, the algorithm computes the optimum value for the objective function and derives coefficients that are used during the

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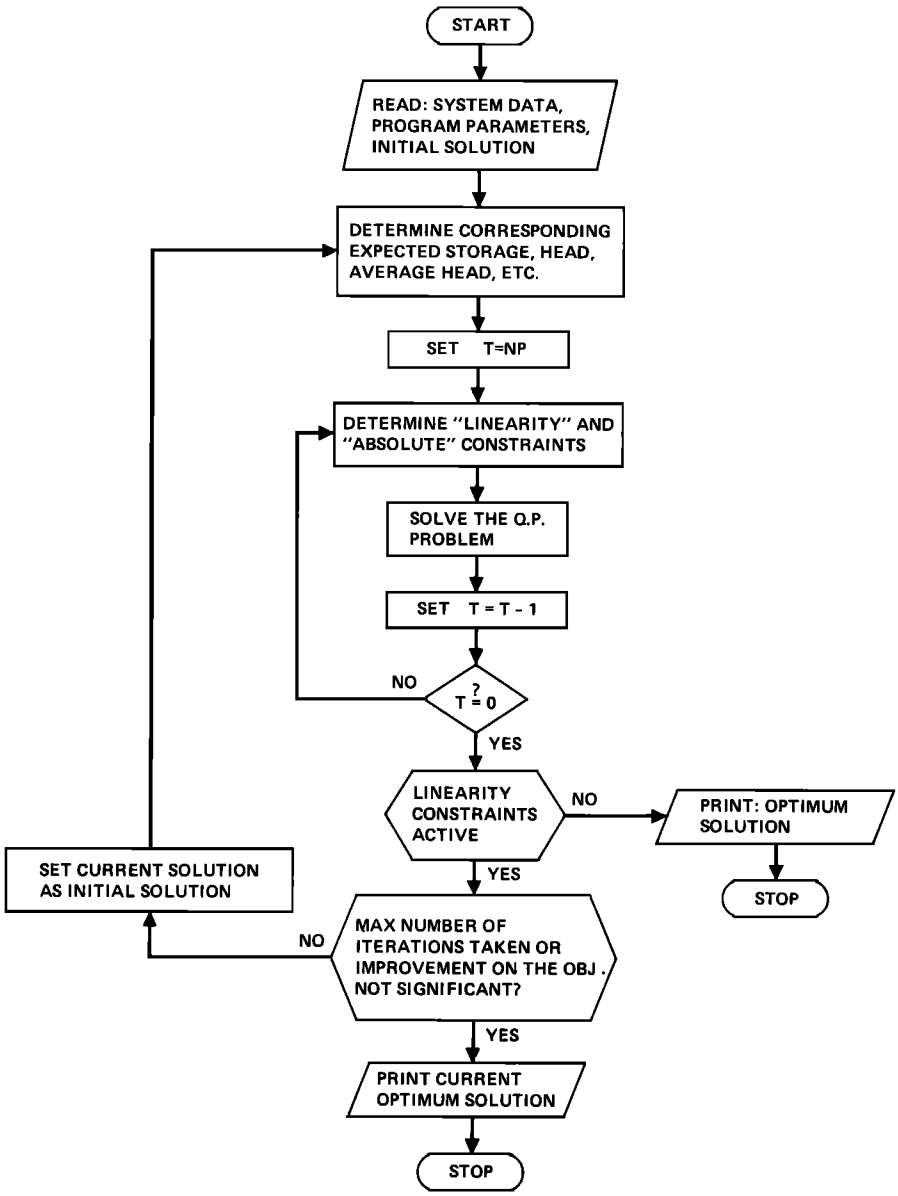


Fig. 3. Flow chart of the proposed algorithm.

forward solution. Thus the optimum values for the control variables can be computed. The entire solution can be achieved in one step or iteration; one step representing the backward-forward solution. It is apparent that if the problem formulation includes bounding constraints, some loss of computational efficiency is expected.

Proposed Algorithm

In order to overcome the limitation imposed by the bounding constraints on state and control variables, we follow the approach of successive approximations. The general idea of the approach is that one starts with an initially feasible solution, but not necessarily optimum, and iterates by continuously improving the current solution. This algorithm terminates whenever the current solution cannot be improved without violating the imposed constraints or, in relation to the number of iterations taken, the improvement of the objective function is considered satisfactory.

For our problem formulation we assume an initial policy $R(t)$, $t = 1, \cdots, NP$. For this policy, we can compute the mean value of the storage vector $E\{S(t)\}$, the mean monthly averaged head $E\{HA(t)\}$, the expected monthly power production, and the expected value of the objective function. If constraints (7) through (9) are satisfied, the initial policy is feasible. With the mean storage computed, the coefficient matrix $A(t)$, the vector $b(t)$ of (5), and the range for which these coefficients are valid can be computed as well. For reference purposes, we call this range "linearity constraints." If the linearity constraints overlap, the upper or lower limits that the operating constraints (7) specify, then the "active" range is adjusted. At each stage of the backward solution, all types of limitations are expressed in terms of constraints on the control variables. This creates an admissible region for the releases $R(t)$.

The proposed algorithm uses an iterative methodology. It starts with an initial "guess" for the optimal problem solution, and during each iteration improves the value of the objective function. The algorithm stops whenever an "exit criterion" is

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met. If, at a given iteration, the optimal point of the quadratic programming problems which were solved for every stage, $t = 1, \cdots, NP$ was found within its admissible region, then the algorithm stops. This is an exit criterion. Under the postulation of concave objective function, global optimum is found; the boundary constraints are not active; the solution of the quadratic programming problems are the optimal problem solution. If for some stage t the optimum of quadratic programming problems is found on a linearity bound, then the solution that is obtained during the current iteration will serve as the initial estimate for the releases for the next iteration. If for all stages $t = 1, \cdots, NP$ no linearity constraints are active, but the quadratic programming optima are confined due to the operating constraints, the algorithm stops. This is again an exit criterion. An optimal problem solution is found; at least one bounding constraint is active. The solution may not be unique, and an alternate optimal solution may have been found if the solution procedure had started at a different initial point. However, the solution cannot be improved without violating the currently active constraints. Figure 3 outlines the basic steps of the developed algorithm.

Considering an initial estimate for the releases enables us to estimate a corresponding "approximate" range for the storages which subsequently permits us to linearize the nonlinear constraint (9) and the generally nonlinear storage-elevation curve. Since the storage-elevation curve is given as a piecewise linear function, we do not need to use Taylor series approximations to linearize it.

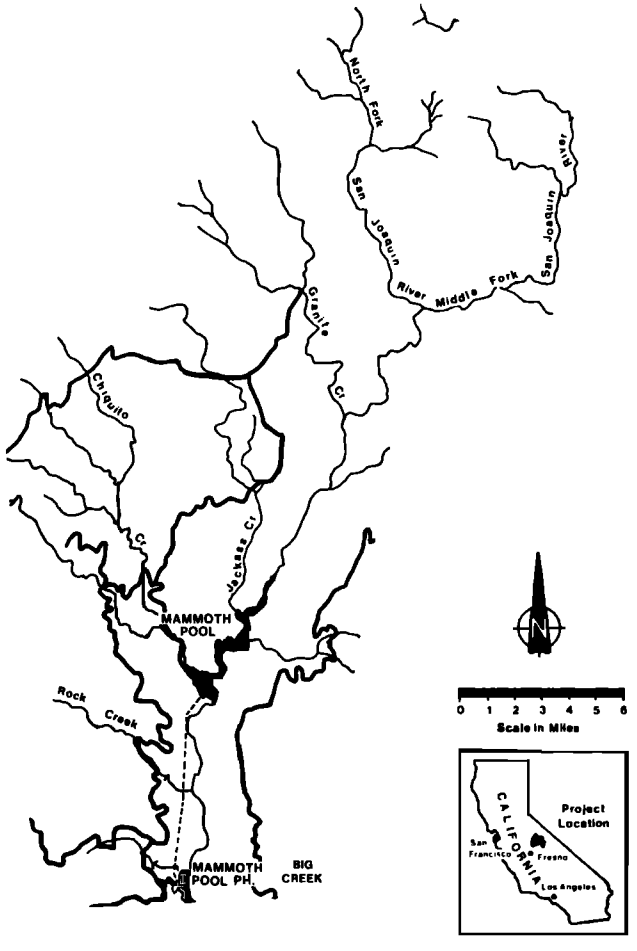


Fig. 4. Upper San Joaquin River basin drainage area (1 mile = 1.609 km).

The proposed algorithm solves the problem of maximizing (16) subject to the system of linear dynamics (13) plus the boundary constraints (7) through (9). For a backward DP, the objective function value can be defined as $V_t(S)$, the maximum energy produced by the system, if the system starts stage (period) t in state (storage) S, and we have not yet learned the actual values of the random variables (inflows) IM at that stage. By induction we can prove that $V_t(S)$ is a quadratic function of S, or

$$V_t(S) = S^T(t) \cdot P(t) \cdot S(t) + q^T(t) \cdot S(t) + z(t)$$

where P(t), q(t), and z(t) are recursive coefficients to be determined during the backward solution. At each stage t a constrained quadratic programming is solved. The quadratic matrix and the cost vector of the objective function, as well as the constraint matrix and the right hand side of the quadratic programming problem, that has to be solved at stage t, are computed based on the values P(t + 1), g(t + 1), and z(t + 1) obtained at the previous stage t + 1, and the current guess for the releases at this stage, R(t). The outcome of this optimization is a new set of estimates for releases for time t. Let RO(t) be the optimum solution of the quadratic programming problem. Assuming for simplicity (though it is not necessary) that there is no cross correlation, it can be shown that

$$P(t) = M3(t) \cdot P(t + 1) \cdot M3(t)$$

$$\begin{aligned} q^T(t) = RO^T(t) \cdot [&M3(t) \cdot A(t + 1) + A(t) + 2 \cdot M3(t) \\ &\cdot P(t + 1) \cdot M5(t)]^T + 2 \cdot QM^T(t) \cdot M4(t) \cdot P(t + 1) \\ &\cdot M3(t) + q(t + 1) \cdot M3(t) \end{aligned}$$

$$\begin{aligned} z(t) = RO^T(t) \cdot [&A(t + 1) \cdot M5^T(t) \cdot P(t + 1) \cdot M5(t)] \cdot RO(t) \\ &+ \{b^T(t) + b^T(t + 1) + QM^T(t) \cdot [M4(t) \cdot A(t + 1) \\ &+ 2 \cdot M4(t) \cdot P(t + 1) \cdot M5(t)] + q^T(t + 1) \cdot M5(t)\} \cdot RO(t) \\ &+ QM^T(t) \cdot M4(t) \cdot P(t + 1) \cdot M4(t) \cdot QM(t) \\ &+ QSD^T(t) \cdot M4(t) \cdot P(t + 1) \cdot M4(t) \cdot QSD(t) \\ &+ g^T(t + 1) \cdot M4(t) \cdot QM(t) + z(t + 1) \end{aligned}$$

where M3(t), M4(t), and M5(t) are matrices defined by

$$M3(t) = [I + \beta(t)]^{-1} \cdot [I - \alpha(t)] \quad (17)$$

$$M4(t) = [I + \beta(t)]^{-1} \cdot [I - \gamma(t)] \quad (18)$$

$$M5(t) = [I + \beta(t)]^{-1} \cdot SYS \quad (19)$$

and QM(t) and QSD(t) are vectors whose elements are the means and standard deviations of the natural inflows. In Appendix A we provide the mathematical development of the new algorithm, we demonstrate how it goes through a complete iteration, and we indicate how it handles the bounding constraints.

Serial Correlation Case

In our analysis so far, we have made the assumption that forecast of inflows is not serially correlated. Implicitly, we have assumed that the forecast results from a conceptual physical model and the uncertainty is due to modeling errors and uncertainties that were given as input to the conceptual

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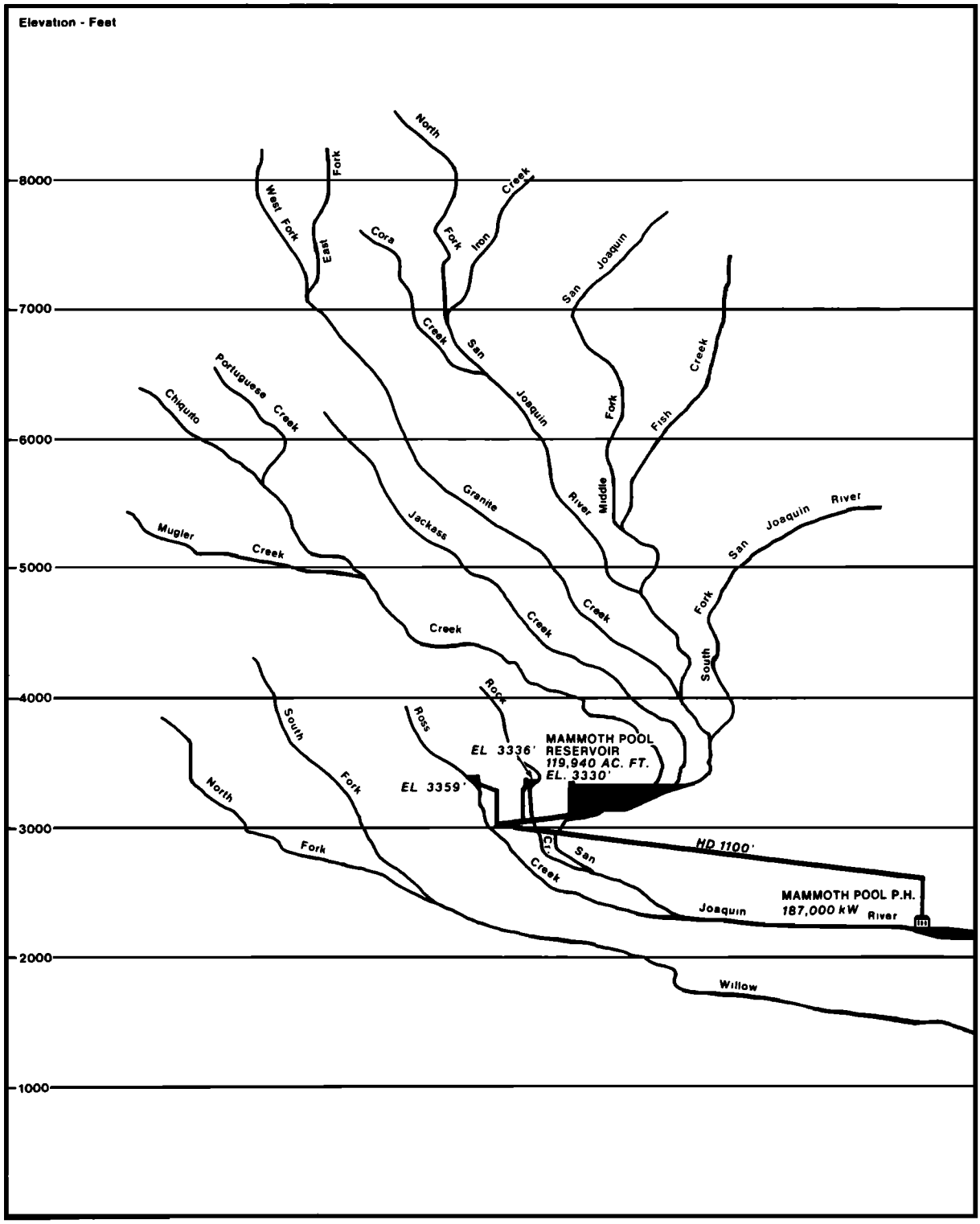


Fig. 5. Schematic profile of San Joaquin River projects (1 foot = 30.48 cm).

model and have been transformed to reservoir inflow variance. Future precipitation is a very large uncertainty.

Here, we examine the required modifications to the proposed solution approach if a statistical model is used for the forecast. In particular, we assume inflows are characterized as an autoregressive process. A similar assumption was made by *Mariño and Loaiciga* [1985], but they used the forecast in a deterministic manner. Here we examine how to implement the specific structure of the forecasting model in our stochastic optimization algorithm.

We assume that inflow is a first-order autoregressive process, or

$$\mathrm{IM}(t) = \mathrm{AR}_t \cdot \mathrm{IM}(t-1) + \varepsilon_t \qquad t = 1, \cdots, \mathrm{NP} \qquad (20)$$

where AR_t is a diagonal matrix containing the first-order autocorrelation coefficients and ε_t is the random component of

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the autocorrelation. Equation (20) can also be written as

$$IM(t) = \left(\prod_{k=1}^{T} AR_K\right) \cdot IM(0) + \left(\prod_{j=i+1}^{T} AR_j \cdot e_i\right) \quad (21)$$

By substituting IM(t) from (21) to (13), the solution can proceed in a similar way as before. The main difference is that the last inflow vector, IM(0) and the autocorrelation coefficients, AR_k, $k = 1, \cdots, NP$ will have an impact on the optimum solution.

CASE STUDIES

To examine the applicability of our research, we have applied the proposed algorithm to two types of case studies. The first was the application of our methodology to the operation of Mammoth reservoir. The reservoir system in this case consisted of a single reservoir. This was chosen for testing purposes since the results obtained analytically can be compared to experimental results obtained by Monte Carlo simulation.

The second type consisted of the application of the proposed algorithm to a system of four reservoirs (NS = 4). For this, the verification of the analytical results by simulation would have been very costly. The purpose of this case study was to demonstrate the applicability of our methodology to large scale systems.

Single-Reservoir Case

Mammoth reservoir is located in the San Joaquin River Basin in central California. It is owned and operated by the Southern California Edison Company. It is part of the company's Big Creek-San Joaquin River hydroelectric development which is on the west side of the Sierra Nevada Mountains. The entire system is a composite of six major reservoirs with nine major tunnels driven through penstock to nine powerhouses.

The Mammoth Pool Dam is the most recent addition to the company's present hydrostorage facilities. It is located across the main San Joaquin River above Big Creek Powerhouse 8.

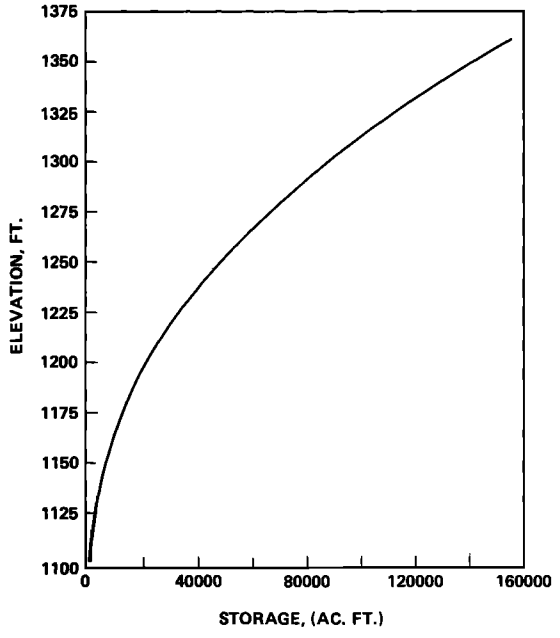


Fig. 6. Storage-elevation curve for Mammoth Lake (1 foot = 30.48 cm; 1 ac-ft = 1233 m^3).

TABLE 1. List of Input Data for the Single Reservoir Case

Period	Minimum Release	Maximum Release	Inflow	Standard Deviation of Inflow
1	4054.	18798.	9801.	6022.
2	10470.	22753.	17354.	4320.
3	17222.	96043.	39670.	32111.
4	18623.	138794.	64035.	46572.
5	40795.	57923.	50950.	12304.
6	46637.	126192.	77775.	29740.
7	85707.	245326.	122798.	68847.
8	124046.	636906.	298349.	211042.
9	63194.	588287.	287269.	229202.
10	16980.	413530.	169175.	181070.
11	8282.	73592.	35215.	31081.
12	4258.	24052.	12481.	8450.

Units are in acre-feet (1 ac-ft = 1233 m^3).

The dam is 820 feet long (1 foot = 30.48 cm), with a net height above stream bed of about 330 feet. The reservoir is approximately eight miles long with a capacity of 119,940 acre-feet (1 ac ft = 1233 m^3). A large part of the runoff of the main San Joaquin River and its western and northern tributary streams, flows into the Mammoth Pool Reservoir for regulation, storage, and use. In addition to producing power at the Mammoth Pool Powerhouse, the water is reused for the same purpose at Big Creek Powerhouses 3 and 4 further downstream. A continuous flow of water is maintained below the dam in order to preserve the fish life in the stream. To recover the energy from such releases, a 1,250-kW unattended generator was built at the end of a power tunnel diversion located at the base of the dam. At the present time the Mammoth Pool Powerhouse has two generating units, each with an operating capacity of 93,500 kW. It is the largest individual plant in the Big Creek development, and is completely remote-controlled. Figures 4 and 5 show the location of Mammoth Lake reservoir relative to the entire San Joaquin reservoir system. Figure 6 shows the storage-elevation curve for the site.

Application and Testing of the Proposed Algorithm

The applicability and validity of the proposed algorithm was tested by the following measures.

1. As the variance of the stochastic variables approaches zero the hydropower production derived by our algorithm should increase. A situation with no variance (deterministic case) corresponds to the case of "perfect" information.
2. As the head of a site increases, the solution should become more "conservative," in the sense of spill becoming more critical.
3. The objective function value as computed by the optimization model must be close to the mean of a sufficient number of Monte Carlo simulation runs.

We used 15 years (1966–1980) of historical data for Mammoth reservoir, as they were provided by Southern California Edison. The data included inflow, release, storage, diversion, and spill for each month. On the basis of the available information we computed estimates for the coefficients $\alpha(t)$, $\beta(t)$, and $\gamma(t)$ of (6). The values used in this case were 0.103, 0.092, and 0.156, respectively.

Mammoth Lake is operated in conjunction with the entire San Joaquin hydro system, and the utility company did not provide us with minimum energy production for the site. We

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also did not have figures for the minimum and maximum releases. We used the power house capacity with the pool 75% full as the maximum allowed production and no minimum production required. We also used the minimum and maximum historical releases as the minimum required and maximum allowed releases, respectively. Table 1 shows the values used in this case.

The algorithm was developed with the assumption that it will be given updated forecast information. Normally, one should expect that the forecast variance should be smaller for the first month than the second, etc. For the purpose of this model application and testing, we used as forecast the historical inflow of the first year and assumed several coefficients of variation, CVR. Table 1 also shows the historical inflow of the first year and the standard deviation of the historical data.

Figure 7 shows the convergence of the algorithm for three different coefficients of variation: CVR = 0., CVR = 0.01, and CVR = 1. As was expected, the smaller the coefficent of variation the higher the objective function value. The number of iterations required for the algorithm to converge also differed. During the model application we noticed that, the algorithm was going through iterations without significant increase of the objective function. For practical purposes, we arbitrarily assigned the algorithm to stop if there was no more than an 0.1% increase in the objective function for more than five consecutive iterations.

For each case we made 100 Monte Carlo simulation runs and computed the mean objective value. The results of the simulation were comparable to the results of the optimization model. The Monte Carlo simulation outcome is also shown on Figure 7. Given the results of Monte Carlo simulation, a probability distribution of the objective function can be estimated.

As is shown in Figure 6, the site has a large elevation difference of 1100 feet between the forebay, when the reservoir is nearly empty, and the afterbay. We changed this to 0% and 50% of the actual amount. An elevation difference of 0 feet corresponds to the case where the power house is at the foot of the dam.

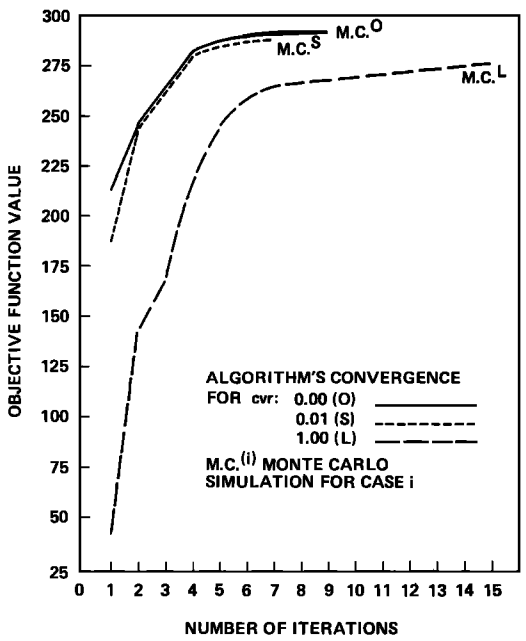


Fig. 7. Sensitivity analysis for input variance.

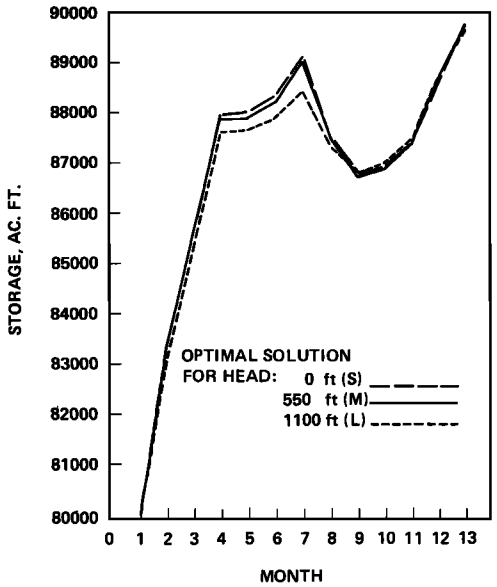


Fig. 8. Sensitivity analysis for elevation head (1 foot = 30.48 cm; 1 ac-ft = 1233 m^3).

Figure 8 shows the mean storages which result by implementing the optimum releases for the three different head drops. All three cases in Figure 8 have the same inflow variance equal to the observed historical variance and the same initial storage.

In all cases we used the expected mean reservoir flows as initial estimates of the releases. The required number of iterations ranged between 10 and 15. The average computer time required for the run of the optimization model was 30 s (IBM 3090).

For testing purposes, we started the algorithm using initial estimates of the releases different than the expected mean reservoir flows. In all the cases the algorithm converged within $\pm 0.2\%$ of the same optimal function value.

Large-Scale System

The main advantage of the new algorithm is its applicability to large scale systems. This advantage emanates from the fact that although we use a DP approach, we do not discretize the state and contral variables. Therefore the computational effort does not increase exponentially with the number of states.

For testing purposes, we applied the proposal model to a hypothetical system of four reservoirs, similar to the one shown in Figure 1*a*. The matrix SYS of (3) is given in this case as

$$\mathrm{SYS} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & -1 \end{bmatrix}$$

for the hypothetical system. Therefore each reservoir had 25, 50, 75, and 100% of Mammoth's head drop and monthly inflows equal to the first 4 years of the given historical data.

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We assumed, although it was not required, that no cross-correlation exists between inflows for each site.

We made several computer runs varying the coefficient of variation for each site, as well as the initial storage. The algorithm converged within 10–20 iterations. The solutions were comparable even when we started from different initial policies. The average computer time required for this case was 1 min (IBM 3090).

On-Peak/Off-Peak Price Difference

In a comparable approach to *Becker and Yeh* [1974], we examined the situation of a given price ratio between on- and off-peak hydroelectric energy production. In this case, the objective function is given by (10) and constraints (11) and (12) should be added to the formulation.

The size of the consequent quadratic programming problems that are required to be solved at each stage of the backward DP solution is essentially doubled. The problem has twice the number of decision variables and a considerably larger number of constraints than the case of no price difference. Note that the exact number of additional constraints would depend on the specific application. Here we used constraints (11) and (12) as the minimum set of constraints to model the enhanced system. Nevertheless, the computational efficiency increases by a factor of 2 and the solution of the quadratic problems does not cause any difficulties.

Figure 9 shows the convergence patterns for different price ratios. In each case we assumed that the price ratio p(t) ws constant for each period t. The same values PEAK(t) as for (12) were used for each case. A price ratio less than or equal to 1 produces essentially the same solution, representing that all the energy can be produced during off-peak hours. In this case, there is no additional benefit in producing hydroelectric energy during certain times of the day. The objective function values in Figures 7 and 9 differ as the former measures the total energy produced, while the latter expresses the net worth of the total energy produced.

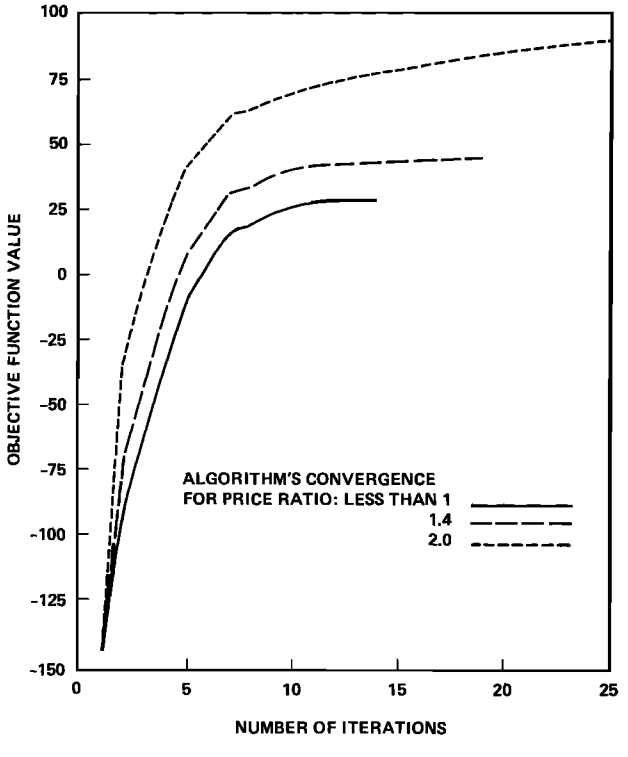


Fig. 9. Sensitivity analysis for energy price ratio.

CONCLUSIONS AND RECOMMENDATIONS

A contribution of this research to the area of water resources has been the extension of constrained DP to the area of stochastic reservoir operation. Depending upon the specific system and the validity of the required assumptions with regard to the system under consideration, the proposed model may become useful in the planning of operating strategies over a specified period. In addition, it may be useful for water resources development, for new facilities planning and for determining maintenance schedules of hydro units.

The proposed algorithm is particularly suited for reservoirs that have a considerably high elevation change for a unit change in storage, rather than reservoirs that have a relatively flat storage-elevation curve. Nevertheless, an optimum operating policy should balance the gain obtained through a higher operating head versus the losses due to spill. However, for reservoirs with flat storage-elevations curves, the gain by maintaining the reservoir at a higher operating head may be minimal.

The existing literature to date has emphasized the need to develop a stochastic optimization model and has mentioned the potential problems of such models. This research aims to be a first step in the direction of using stochastic DP techniques in water resources planning and management.

On the basis of the experience we gained, we see the following topics as natural extensions of our work.

1. Develop a stochastic optimization model for daily operation with the results of our algorithm as input to these models. A problem that may be encountered in this extension is the unavoidable input correlation from period to period.
2. Develop expressions for the probability distribution of the objective function. Since the inflow is a random variable and it is observed after the decision is made, the objective function is actually a controllable and dependent random variable. The proposed algorithm maximizes the mean of the objective function, but ignores higher moments of its distribution. The variance of the objective function may be used as an objective to be minimized as it is discussed by *Trezos* [1982]. A potential problem that we foresee is the difficulty in developing closed form expressions for the higher moments.
3. Since the decision for reservoir operation is sequential in a stochastic environment, it should be of interest to examine how a decision made at an initial stage would restrict or broaden the feasible region of subsequent decisions. This extension may require discretization of the input random variables and cause dimensionality in its application.

Our concluding remarks are as follows. (1) Avoiding discretization of the state and control variables in a DP formulation is a very powerful technique; it allows the model to be applicable to large-scale systems. (2) Much has been done in the area of deterministic optimization of reservoir operation. There is immense room for research in the area of stochastic optimization of water resources systems.

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APPENDIX: MATHEMATICAL DEVELOPMENT OF THE NEW ALGORITHM

The objective is to

$$\max_{CV}\ E_{RV}\left\{\frac{1}{2}\cdot\sum_{t=1}^{NP} S^T(t)\cdot[A(t)+[I-\alpha(t)]\cdot[I+\beta(t)]^{-1}\cdot A(t+1)]\right.$$
$$\cdot R(t)+R^T(t)\cdot SYS^T\cdot[I+\beta(t)]^{-1}\cdot A(t+1)\cdot R(t)$$
$$+[IM^T(t)\cdot[I-\gamma(t)]\cdot[I+\beta(t)]^{-1}\cdot A(t+1)+b^T(t)+b^T(t+1)]$$
$$\cdot R(t)+S^T(NP+1)\cdot PEN\cdot S(NP+1)$$
$$\left.-2\cdot TARG^T(NP+1)\cdot PEN\cdot S(NP+1)+z\right\} \quad (A1)$$

The algorithm requires an initial feasible estimate of the solution R(t), t = 1, ···, NP. Based on this, we can compute the expected storage for each period S(t), t = 2, ···, NP + 1, average head, HA(t), and the expected production, HA(t) · R(t), t = 1, ···, NP. If constraints (7), (8), and (9) are satisfied, the initial estimate is feasible. In addition to checking for the feasibility of the initial estimate, we use the computed estimates of storage to linearize the nonlinear constraint (9).

Based on the definition of the objective function value $V_t(S)$ (the maximum energy produced by the system, if the system starts stage (period) t in state (storage) S and we have not yet learned the actual values of the random variables (inflows) IM at that stage) it is clear that

$$V_{NP+1}[S(NP+1)]=S^T(NP+1)\cdot PEN\cdot S(NP+1)$$
$$-2\cdot TARG^T(NP+1)\cdot PEN\cdot S(NP+1)+z$$

Setting P(NP + 1) = PEN, $q^T(NP+1) = -2\cdot TARG^T(NP+1)\cdot PEN$ and z(NP + 1) = z then we can have

$$V_{NP+1}[S(NP+1)]=S^T(NP+1)\cdot P(NP+1)\cdot S(NP+1)$$
$$+q^T(NP+1)\cdot S(NP+1)+z(NP+1) \quad (A2)$$

By substituting the expressions for M3, M4, and M5 from (17), (18), and (19) into (13), and setting t equal to NP, we have

$$S(NP+1)=M3(NP)\cdot S(NP)$$
$$+M4\cdot R(NP)+M5(NP)\cdot IM(NP) \quad (A3)$$

We now substitute S(NP + 1) from (A3) into (A2) and we obtain $V_{NP+1}[S(NP+1)]$ as a function of S(NP) and R(NP). To simplify the notation, we drop the stage index if it is NP and then (A2) becomes

$$V_{NP+1}[S(NP+1)]=S^T\cdot M3\cdot P(NP+1)\cdot M3\cdot S$$
$$+S^T\cdot M3\cdot P(NP+1)\cdot M4\cdot IM+S^T\cdot M3\cdot P(NP+1)$$
$$\cdot M5\cdot R+IM^T\cdot M4\cdot P(NP+1)\cdot M3\cdot S+IM^T\cdot M4$$
$$\cdot P(NP+1)\cdot M4\cdot IM+IM^T\cdot M4\cdot P(NP+1)\cdot M5\cdot R$$
$$+R^T\cdot M5\cdot P(NP+1)\cdot M3\cdot 5+R^T\cdot M5^T\cdot P(NP+1)$$
$$\cdot M4\cdot IM+R^T\cdot M5^T\cdot P(NP+1)\cdot M5\cdot R \quad (A4)$$

Moving backwards one stage, we compute $V_{NP}(S)$ as

$$V_{NP}(S)=\max_{R(NP)}\ E_{RV}\{S^T(NP)\cdot[A(NP)+[I-\alpha(NP)]$$
$$\cdot[I+\beta(NP)]^{-1}\cdot A(NP+1)]\cdot R(NP)+R^T(NP)\cdot SYS^T$$
$$\cdot[I+\beta(NP)]^{-1}\cdot A(NP+1)\cdot R(NP)+[IM^T(NP)$$
$$\cdot[I-\gamma(NP)]\cdot[I+\beta(NP)]^{-1}\cdot A(NP+1)+b^T(NP)$$
$$+b^T(NP+1)]\cdot R(NP)+V_{NP+1}[S(NP+1)]\} \quad (A5)$$

Substituting $V_{NP+1}[S(NP+1)]$ of (A5) using (A4) we obtain

$$V_{NP}(S)=\max_{R}E\{R^T\cdot[A(NP+1)\cdot M5+M5^T\cdot P(NP+1)\cdot M5]$$
$$\cdot R+[S^T\cdot[M3\cdot A(NP+1)+A+2\cdot M3\cdot P(NP+1)\cdot M5$$
$$+b^T+b^T(NP+1)+IM^T\cdot[M4\cdot A(NP+1)+2\cdot M4$$
$$\cdot P(NP+1)\cdot M5]+q^T(NP+1)\cdot M5]\cdot R+S^T\cdot M3$$
$$\cdot P(NP+1)\cdot M3\cdot S+2\cdot IM^T\cdot M4\cdot P(NP+1)\cdot M3\cdot S$$
$$+IM^T\cdot M4\cdot P(NP+1)\cdot M4\cdot IM+q^T(NP+1)\cdot M3\cdot S$$
$$+q^T(NP+1)\cdot M4\cdot IM+z(NP+1)\} \quad (A6)$$

We now take the expectation of the argument within the brackets and (A6) becomes

$$V_{NP}(S)=\max_{R}\{R^T\cdot[A(NP+1)\cdot M5+M5^T\cdot P(NP+1)\cdot M5]$$
$$\cdot R+[S^T\cdot[M3\cdot A(NP+1)+A+2\cdot M3\cdot P(NP+1)\cdot M5]$$
$$+b^T+b^T(NP+1)+QM^T\cdot[M4\cdot A(NP+1)+2\cdot M4$$
$$\cdot P(NP+1)\cdot M5]+q^T(NP+1)\cdot M5]$$
$$\cdot R+S^T\cdot M3\cdot P(NP+1)\cdot M3\cdot S+2\cdot QM^T\cdot M4\cdot P(NP+1)$$
$$\cdot M3\cdot S+QM^T\cdot M4\cdot P(NP+1)\cdot M4\cdot QM+QSD^T$$
$$\cdot M4\cdot P(NP+1)\cdot M4\cdot QSD+q^T(NP+1)\cdot M3$$
$$\cdot S+q^T(NP+1)\cdot M4\cdot QM+z(NP+1)\} \quad (A7)$$

The above expression for $V_{NP}(S)$ is the optimum of a quadratic function for R. Since R is constrained, a quadratic programming routine is used at this point to determine where the above function of R achieves its maximum.

The quadratic matrix of the quadratic programming problem is

$$DM=A(NP+1)\cdot M5+M5^T\cdot P(NP+1)\cdot M5 \quad (A8)$$

The transpose of the cost vector for the quadratic programming problem is

$$cv^T=S^T\cdot[M3\cdot A(NP+1)+A+2\cdot M3P(NP+1)\cdot M5]$$
$$+b^T+b^T(NP+1)+QM^T\cdot[M4\cdot A(NP+1)$$
$$+2\cdot M4\cdot P(NP+1)\cdot M5]+q^T(NP+1)\cdot M5 \quad (A9)$$

Since the objective function is to be maximized, DM must be negative semidefinite in order to guarantee optimality.

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The constraints of the quadratic programming problem are

$$SLL(NP+1) \le E\{S(NP+1)\} \le SUL(NP+1) \quad (A10)$$

$$RLL(NP+1) \le R(NP+1) \le RUL(NP+1) \quad (A11)$$

$$CONTR(NP) \le E\{HA(NP)\cdot R(NP)\} \le SUL(NP+1) \quad (A12)$$

Substituting A3 into A10 we have

$$SLL(NP+1) \le E\{M3\cdot S+M4\cdot R+M5\cdot IM\} \le SUL(NP+1)$$

Making the approximation that E{S(NP)} is equal to the current estimate of S(NP), we develop a deterministic constraint on R(NP):

$$SLL(NP+1) - M3\cdot S - M5\cdot QM \le M4\cdot R$$

$$R \le SUL(NP+1) - M3\cdot S - M5\cdot QM \quad (A13)$$

Similarly, we could substitute (A3) into (14) and the resulting expression for HA(NP) into (A12) but this approach would have resulted in a nonlinear constraint for R(NP). Instead, we use the current estimates for S(NP) and S(NP + 1) to approximate HA(NP) and then (A12) becomes

$$CONTR(NP) \le HA(NP)\cdot R(NP) \le CAPM(NP) \quad (A14)$$

Therefore the quadratic programming problem is to maximize (A7) subject to (A11), (A13), and (A14).

Let RO(NP) be the optimum solution to the quadratic programming problem. Then from (A7) we conclude that

$$\begin{aligned} V_{NP}(S) = {} & RO^T\cdot[A(NP+1)\cdot M5+M5^T\cdot P(NP+1)\cdot M5]\cdot RO \\ & + [S^T\cdot[M3\cdot A(NP+1)+A+2\cdot M3\cdot P(NP+1)\cdot M5] \\ & + b^T + b^T(NP+1) + QM^T\cdot[M4\cdot A(NP+1)+2\cdot M4 \\ & \cdot P(NP+1)\cdot M5] + q^T(NP+1)\cdot M5]\cdot RO + S^T\cdot M3 \\ & \cdot P(NP+1)\cdot M3\cdot S + 2\cdot QM^T\cdot M4\cdot P(NP+1)\cdot M3\cdot S \\ & + QM^T\cdot M4\cdot P(NP+1)\cdot M4\cdot QM + QSD^T\cdot M4 \\ & \cdot P(NP+1)\cdot M4\cdot QSD + q^T(NP+1)\cdot M3\cdot S \\ & + q^T(NP+1)\cdot M4\cdot QM + z(NP+1) \quad (A15) \end{aligned}$$

which can be rearranged as

$$\begin{aligned} V_{NP}(S) = {} & S^T\cdot M3\cdot P(NP+1)\cdot M3\cdot S + S^T\cdot[M3\cdot A(NP+1) \\ & + A + 2\cdot M3\cdot P(NP+1)\cdot M5]\cdot RO + 2\cdot QM^T\cdot M4 \\ & \cdot P(NP+1)\cdot M3\cdot S + q(NP+1)\cdot M3\cdot S + RO^T\cdot[A(NP+1) \\ & \cdot M5 + M5^T\cdot P(NP+1)\cdot M5\cdot RO + \{b^T + b^T(NP+1) \\ & + QM^T\cdot[M4\cdot A(NP+1)+2\cdot M4\cdot P(NP+1)\cdot M5] \\ & + q^T(NP+1)\cdot M5\}\cdot RO + QM^T\cdot M4\cdot P(NP+1)\cdot M4\cdot QM \\ & + QSD^T\cdot M4\cdot P(NP+1)\cdot M4\cdot QSD \\ & + q^T(NP+1)\cdot M4\cdot QM + z(NP+1) \quad (A15) \end{aligned}$$

By setting

$$P(NP) = M3\cdot P(NP+1)\cdot M3$$

$$\begin{aligned} q^T(NP) = {} & RO^T \\ & \cdot[M3\cdot A(NP+1)+A+2\cdot M3\cdot P(NP+1)\cdot M5]^T \\ & + 2\cdot QM^T\cdot M4\cdot P(NP+1)\cdot M3 + q(NP+1)\cdot M3 \end{aligned}$$

$$\begin{aligned} z(NP) = {} & RO^T\cdot[A(NP+1)\cdot M5 + M5^T\cdot P(NP+1)\cdot M5] \\ & \cdot RO + \{b^T + b^T(NP+1) + QM^T\cdot[M4\cdot A(NP+1) \\ & + 2\cdot M4\cdot P(NP+1)\cdot M5] + q^T(NP+1)\cdot M5\}RO \\ & + QM^T\cdot M4\cdot P(NP+1)\cdot M4\cdot QM + QSD^t\cdot M4\cdot P(NP+1) \\ & \cdot M4\cdot QSD + q^T(NP+1)\cdot M4\cdot QM + z(NP+1) \end{aligned}$$

We have completed one step in the backward DP solution. We now can compute RO(NP − 1), P(NP − 1), q(NP − 1), and z(NP − 1), etc. At the end of the backward solution we would have new estimates for the optimal solution RO(t), t = 1, ···, NP. This concludes a complete iteration of the algorithm.

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